12 Graph Minors

Our goal in this last chapter is a single theorem, one which dwarfs any other result in graph theory and may doubtless be counted among the deepest theorems that mathematics has to offer: *in every infinite set of graphs there are two such that one is a minor of the other.* This *graph minor theorem*, inconspicuous though it may look at first glance, has made a fundamental impact both outside graph theory and within. Its proof, due to Neil Robertson and Paul Seymour, takes well over 500 pages.

So we have to be modest: of the actual proof of the graph minor theorem this chapter will convey only a very rough impression. However, as with most truly fundamental results, the proof has sparked off the development of methods of quite independent interest and potential. This is true particularly for the use of *tree-decompositions*, a concept that is not only central to graph minor theory but has found algorithmic applications too, and *tangles*, a radically new notion of high connectivity somewhere inside a given graph.

Section 12.1 gives an introduction to *well-quasi-ordering*, a concept central to the graph minor theorem. In Section 12.2 we apply this to prove the graph minor theorem for trees. We study tree-decompositions in Sections 12.3–12.4, and tangles in Section 12.5. In Section 12.6 we look at *forbidden-minor theorems*: results in the spirit of Kuratowski's theorem (4.4.6) or Wagner's theorem (7.3.4), which describe the structure of the graphs not containing some specified graph or graphs as a minor.

In Section 12.7 we give a direct proof of the 'generalized Kuratowski' theorem that embeddability in any fixed surface can be characterized by forbidding finitely many minors. We conclude with an overview of the proof and implications of the graph minor theorem itself.

R. Diestel, *Graph Theory*, Graduate Texts in Mathematics 173,
https://doi.org/10.1007/978-3-662-70107-2_12

12.1 Well-quasi-ordering

$\leqslant, <$ A reflexive and transitive binary relation $\leqslant$ on a set X is a *quasi-ordering.* We write $x < y$ for '$x \leqslant y$ and $y \not\leqslant x$'. A quasi-ordering $\leqslant$ on X is a *well-quasi-ordering* (margin: *well-quasi-ordering*), and the elements of X are *well-quasi-ordered* by $\leqslant$, if for every infinite sequence $x_0, x_1, \ldots$ in X there are indices $i < j$ such that $x_i \leqslant x_j$. Then (x_i, x_j) is a *good pair* (margin: *good pair*) of this sequence. A sequence containing a good pair is a *good sequence*; thus, a quasi-ordering on X is a well-quasi-ordering if and only if every infinite sequence in X is good. An infinite sequence is *bad* (margin: *good/bad sequence*) if it is not good.

Proposition 12.1.1. *A quasi-ordering $\leqslant$ on X is a well-quasi-ordering if and only if X contains neither an infinite antichain nor an infinite strictly decreasing sequence $x_0 > x_1 > \ldots$.*

(9.1.2) *Proof.* The forward implication is trivial. Conversely, let $x_0, x_1, \ldots$ be any infinite sequence in X. Let K be the complete graph on $\mathbb{N} = \{0, 1, \ldots\}$. Colour the edges ij $(i < j)$ of K with three colours: green if $x_i \leqslant x_j$, red if $x_i > x_j$, and amber if x_i, x_j are incomparable. By Ramsey's theorem (9.1.2), K has an infinite induced subgraph whose edges all have the same colour. If there is neither an infinite antichain nor an infinite strictly decreasing sequence in X, then this colour must be green, i.e. $x_0, x_1, \ldots$ has an infinite subsequence in which every pair is good. In particular, the sequence $x_0, x_1, \ldots$ is good. □

In the proof of Proposition 12.1.1, we showed more than was needed: rather than finding a single good pair in $x_0, x_1, \ldots$, we found an infinite increasing subsequence. We have thus shown the following:

Corollary 12.1.2. *If X is well-quasi-ordered, then every infinite sequence in X has an infinite increasing subsequence.* □

The following lemma, and the idea of its proof, are fundamental to the theory of well-quasi-ordering. Let $\leqslant$ be a quasi-ordering on a set X.
$\leqslant$ For finite subsets $A, B \subseteq X$, write $A \leqslant B$ if there is an injective mapping $f: A \to B$ such that $a \leqslant f(a)$ for all $a \in A$. This naturally extends $\leqslant$ to
$[X]^{<\omega}$ a quasi-ordering on $[X]^{<\omega}$, the set of all finite subsets of X.

[12.2.1] [12.7.1] **Lemma 12.1.3.** *If X is well-quasi-ordered by $\leqslant$, then so is $[X]^{<\omega}$.*

Proof. Suppose that $\leqslant$ is a well-quasi-ordering on X but not on $[X]^{<\omega}$. We start by constructing a bad sequence $(A_n)_{n \in \mathbb{N}}$ in $[X]^{<\omega}$, as follows. Given $n \in \mathbb{N}$, assume inductively that A_i has been defined for every $i < n$, and that there exists a bad sequence in $[X]^{<\omega}$ starting with $A_0, \ldots, A_{n-1}$. (This is clearly true for $n = 0$: by assumption, $[X]^{<\omega}$ contains a bad sequence, and this has the empty sequence as an initial segment.) Choose $A_n \in [X]^{<\omega}$ so that some bad sequence in $[X]^{<\omega}$ starts with $A_0, \dots, A_n$ and $|A_n|$ is as small as possible.

Clearly, $(A_n)_{n \in \mathbb{N}}$ is a bad sequence in $[X]^{<\omega}$; in particular, $A_n \neq \emptyset$ for all n. For each n pick an element $a_n \in A_n$ and set $B_n := A_n \setminus \{a_n\}$.

By Corollary 12.1.2, the sequence $(a_n)_{n \in \mathbb{N}}$ has an infinite increasing subsequence $(a_{n_i})_{i \in \mathbb{N}}$. By the minimal choice of A_{n_0}, the sequence

$$A_0, \dots, A_{n_0-1}, B_{n_0}, B_{n_1}, B_{n_2}, \dots$$

is good; consider a good pair. Since $(A_n)_{n \in \mathbb{N}}$ is bad, this pair cannot have the form (A_i, A_j) or (A_i, B_j), as $B_j \leqslant A_j$. So it has the form (B_i, B_j). Extending the injection $B_i \to B_j$ by $a_i \mapsto a_j$, we deduce again that (A_i, A_j) is good, a contradiction. □

12.2 The graph minor theorem for trees

The graph minor theorem can be expressed by saying that the finite graphs are well-quasi-ordered by the minor relation $\preccurlyeq$. Indeed, by Proposition 12.1.1 and the obvious fact that no strictly descending sequence of minors can be infinite, being well-quasi-ordered is equivalent to the non-existence of an infinite antichain, the formulation used earlier.

In this section, we prove a strong version of the graph minor theorem for trees:

Theorem 12.2.1. (Kruskal 1960) [12.7.1]
The finite trees are well-quasi-ordered by the topological minor relation.

We shall base the proof of Theorem 12.2.1 on the following notion of an embedding between rooted trees, which strengthens the usual embedding as a topological minor. Consider two trees T and T', with roots r and r' say. Let us write $T \leqslant T'$ if there exists an isomorphism φ, from some subdivision of T to a subtree T'' of T', that preserves the tree-order on $V(T)$ associated with T and r. (Thus if $x < y$ in T then $\varphi(x) < \varphi(y)$ in T'; see Fig. 12.2.1.) As one easily checks, this is a quasi-ordering on the class of all rooted trees. $T \leqslant T'$

Proof of Theorem 12.2.1. We show that the rooted trees are well-quasi-ordered by the relation $\leqslant$ defined above; this clearly implies the theorem. (12.1.3)

Suppose not. To derive a contradiction, we proceed as in the proof of Lemma 12.1.3. Given $n \in \mathbb{N}$, assume inductively that we have chosen a sequence $T_0, \dots, T_{n-1}$ of rooted trees such that some bad sequence of rooted trees starts with this sequence. Choose as T_n a minimum-order T_n